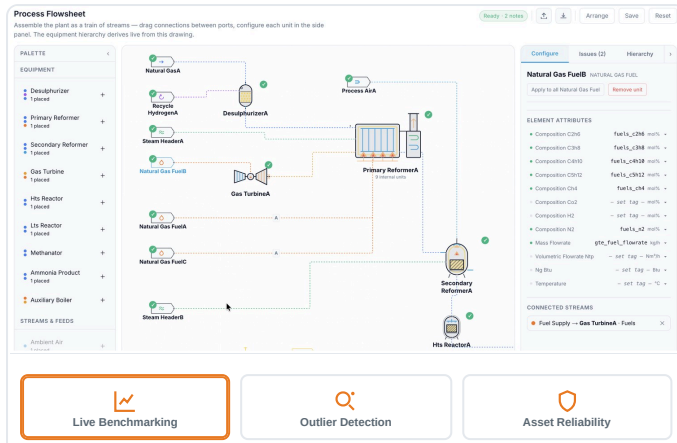


## PRODUCT OVERVIEW

ammoniaX360 is an AI powered decision support application that turns ammonia plant data into measurable performance opportunities. Domain and machine learning agents benchmark the reforming train, shift conversion, CO<sub>2</sub> removal and synthesis loop, delivering ranked, quantified recommendations across energy, conversion, purge and equipment performance. Every opportunity links to its driving parameter for transparent root cause insight. Configured and commissioned by Ingenero process engineers to reflect each plant's feed and operating envelope.

### 01 Unit Configuration

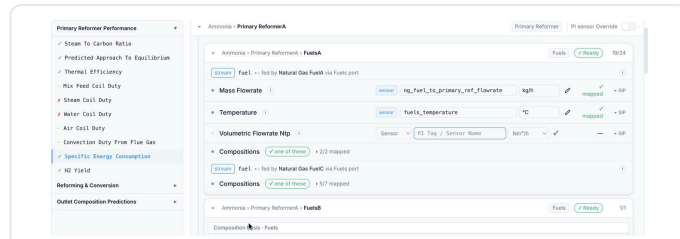
Shown: plant configuration, then model selection against the assets.



|                                 |                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Asset hierarchy</b>          | feed treatment, primary & secondary reformer, HT/LT shift, CO <sub>2</sub> removal, methanation, synthesis loop |
| <b>Reformer &amp; converter</b> | reformer radiation and convection arrangement, converter beds and quench points                                 |
| <b>Product streams</b>          | make-up gas, purge gas, CO <sub>2</sub> product spec                                                            |

### 02 Plant Data and Model Execution

Shown: KPI packages with sensor input workflow.



|                                                                |                                                                        |                                                               |
|----------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>AMMONIA PRODUCTION</b><br>(MTPD)<br><b>1,842</b>            | <b>SYNTHESIS CONVERSION</b><br>(%/PASS)<br><b>26.4</b>   <b>27.5</b>   | <b>SPECIFIC ENERGY</b><br>(GJ/T)<br><b>32.1</b>   <b>30.5</b> |
| <b>STEAM TO CARBON</b><br>(MOL/MOL)<br><b>3.2</b>   <b>2.9</b> | <b>REFORMER EXCESS O<sub>2</sub></b><br>(%)<br><b>3.1</b>   <b>2.0</b> | <b>LOOP INERTS</b><br>(MOL %)<br><b>11.8</b>   <b>10.0</b>    |

|                   |                                                                                 |
|-------------------|---------------------------------------------------------------------------------|
| <b>Inputs</b>     | Sensor tag entry along with design tag values with configuration details        |
| <b>Processing</b> | tag validation, tag health and stuck-signal checks, outlier and range screening |
| <b>Outputs</b>    | predicted KPIs, ranked actionables with cause, trends, alerts                   |

### 03 AI-Powered Model Agent Suite

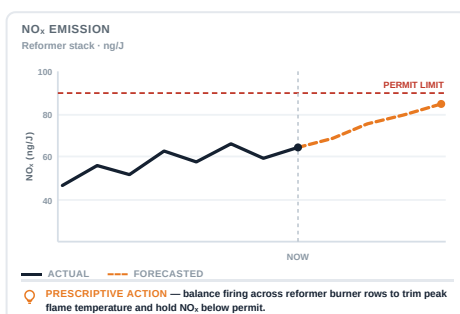
Agent/Module selection suitable to the plant needs.

- Thermal & Fuel Efficiency**  
Cuts fuel loss from excess O<sub>2</sub>, uneven firing and stack losses.
- Environment KPIs**  
Forecasted NO<sub>x</sub> emissions for early excursion alerts.
- GenAI Copilot**  
Answers plant questions inside the operating workflow.
- Reformer Optimizer**  
Balances firing, steam to carbon and severity to cut fuel and steam.
- Synthesis Loop Optimizer**  
Maximizes conversion per pass and manages loop inerts.
- CO<sub>2</sub> Removal Monitor**  
Tracks solvent circulation, CO<sub>2</sub> slip and regeneration energy.

- Digital Analysers**  
Reforming outlet composition prediction, methane slip and approach to equilibrium.
- Catalyst Run Length**  
Real time forecasting of reforming and HT/LT shift catalyst activity to plan the next changeover.
- Boiler Leak Detection**  
Early detection of waste heat boiler tube leaks generating alerts days before complete failure.
- Convection Fouling in Reformer**  
Tracks convection bank coil fouling and heat recovery decline to plan cleaning.
- Purge Recovery**  
Reduces downtime and improves efficiency of membrane and cold box separation units.
- Asset Reliability**  
Outlier detection flags equipment anomalies and failure risk.

### 04 Opportunity and Operator Actions

|                                                                    | BUSINESS KPI                  | CAUSE MESSAGE                             | ACTUAL           | OPTIMUM          | SUGGESTION                                                                                               |
|--------------------------------------------------------------------|-------------------------------|-------------------------------------------|------------------|------------------|----------------------------------------------------------------------------------------------------------|
| <b>PREDICTED PRODUCTION OPPORTUNITY</b><br>(MT/DAY)<br><b>18.6</b> | <b>LOW AMMONIA PRODUCTION</b> | <b>HIGH INERTS IN SYNTHESIS LOOP</b>      | <b>12.5 mol%</b> | <b>10.0 mol%</b> | INCREASE PURGE RATE TO REDUCE LOOP INERTS AND IMPROVE CONVERSION PER PASS.                               |
|                                                                    |                               | <b>LOW SYNTHESIS LOOP PRESSURE</b>        | <b>138 barg</b>  | <b>145 barg</b>  | RAISE SYNTHESIS LOOP PRESSURE TOWARD OPTIMUM TO IMPROVE AMMONIA CONVERSION PER PASS.                     |
| <b>PREDICTED ENERGY OPPORTUNITY</b><br>(MMBTU/DAY)<br><b>276</b>   | <b>HIGH SPECIFIC ENERGY</b>   | <b>HIGH REFORMER EXCESS O<sub>2</sub></b> | <b>3.1 %</b>     | <b>2.0 %</b>     | REDUCE EXCESS O <sub>2</sub> TO OPTIMUM TO LOWER FUEL GAS CONSUMPTION WHILE MAINTAINING SAFE COMBUSTION. |
|                                                                    |                               | <b>HIGH STEAM TO CARBON RATIO</b>         | <b>3.2</b>       | <b>2.9</b>       | REDUCE STEAM TO CARBON RATIO TO OPTIMUM TO CUT REFORMER STEAM AND FUEL USE.                              |



|                                                                                                               |                                                                                                                          |                                                                                                     |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>1–3%</b> <b>Ammonia production increase</b><br>Synthesis loop optimization and improved reforming severity | <b>2–5%</b> <b>Specific energy reduction</b><br>Excess O <sub>2</sub> , steam to carbon and reformer firing optimization | <b>5–15%</b> <b>Reduction in losses</b><br>Lower solvent, purge and hydrogen losses across the loop |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|

### 05 Platform

|                                                                                  |                                                                                              |                                                                              |                                                                           |                                                                                 |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Historian connectivity</b><br>connects directly with leading plant historians | <b>Simulation integration</b><br>third-party simulators integrate without custom engineering | <b>Daily reporting</b><br>downloadable daily reports with interactive trends | <b>GenAI Copilot</b><br>answers plant questions in the operating workflow | <b>Build your own dashboard</b><br>engineers assemble the views each unit needs |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|